Lecture 20: Singular Value Decomposition

Balls, Gaussian Clouds, and Real Orthogonal Axes

The Visual Question

Today's Question

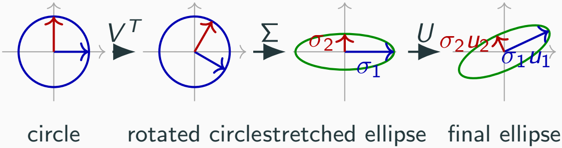
$$\underbrace{x \in \mathbb{R}^n}_{\text{input vector}} \xrightarrow[]{\underbrace{A}_{m\times n \text{ real matrix}}} \underbrace{Ax \in \mathbb{R}^m}_{\text{output vector}}.$$

Instead of asking for eigenvectors of A, ask a geometric question:

What does A do to the round unit ball?

SVD is the axis system of that transformed shape.

The Picture First: Circle to Ellipse



The formula is visual: circle $\rightarrow$ circle $\rightarrow$ ellipse $\rightarrow$ rotated ellipse.

Our Goal After the Picture

$$A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{final output axes}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{axis lengths}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{special input coordinates}}}.$$

The picture says: rotate to special coordinates, stretch, rotate back.

The Three Actions

$$A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{rotate again}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{stretch}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{rotate}}}.$$

$$\boxed{\text{input vector}} \xrightarrow{V^T} \boxed{\text{rotate by } V^T} \xrightarrow{\Sigma} \boxed{\text{stretch by } \Sigma} \xrightarrow{U} \boxed{\text{rotate again by } U}$$

Rotate, stretch, rotate again.

Why $A^T A$ Appears

Reverse-Engineer the Method

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}}.$$

$$\underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix} = I}_{U^TU=I} \quad \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} = I}_{V^TV=I}.$$

$$\underbrace{A^T}_{\text{transpose of the contract}} = \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T \text{ in the square-picture notation}} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T}.$$

$$\underbrace{A^TA}_{n\times n \text{ input-side square}}$$

$$\underbrace{AA^T}_{m\times m \text{ output-side square}} \quad .$$

Input Axes Come from $A^T A$

$$A^T A = \left(\underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T} \right) \left(\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \right)$$

$$= \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{I} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}$$

$$= \boxed{\underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T \Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}}.$$

$$\underbrace{\Sigma^T \Sigma}_{\text{squared singular values}} = \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\text{diagonal matrix}}.$$

V is an orthogonal eigenbasis of $A^T A$.

Output Axes Come from AA^T

$$AA^T = \left(\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \right) \left(\underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T} \right)$$

$$= \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{I} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T}$$

$$= \boxed{\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma\Sigma^T} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T}}.$$

$$\underbrace{\Sigma\Sigma^T}_{\text{squared singular values}} = \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\text{diagonal matrix}}.$$

U is an orthogonal eigenbasis of AA^T.

The Method

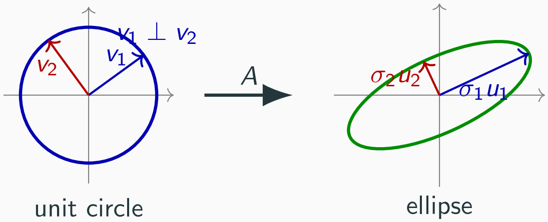
$$A^T A = \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{\substack{V \\ \text{input axes } v_i}} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma^T \Sigma \\ \text{squared lengths } \sigma_i^2}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T},$$

$$AA^T = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{output axes } u_i}} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \Sigma^T \\ \text{squared lengths } \sigma_i^2}} \underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T}.$$

The square matrices reveal the axes; the diagonal entries reveal the lengths.

Axes of the Ellipse

Circle to Ellipse



The image of a round object has special axes.

Contract First

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{columns define } u_i}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{diagonal entries define } \sigma_i}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{rows define } v_i^T}} .}$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \quad \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \quad \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} = I}_{V^T V = I} .$$

Contract Gives the Axis Equation

$$
A\underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \left(\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \right) \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V}
$$

$$
= \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{I}
$$

$$
= \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma}.
$$

$$
\boxed{AV = U\Sigma.}
$$

Expand AV

$$\underbrace{A \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{\substack{\text{special input directions} \\ V}} = \underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{\text{left side columns}}.$$

Expand $U\Sigma$

$$\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{output directions}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{diagonal stretches}}} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0 u_n \\ | & | & & | \end{pmatrix}}_{\text{right side columns}}.$$

Compare Columns

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0 u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma}.$$

$$\underbrace{Av_1 = \sigma_1 u_1}_{\text{first column}}, \quad \underbrace{Av_2 = \sigma_2 u_2}_{\text{second column}}, \quad \cdots, \quad \underbrace{Av_n = 0 u_n}_{\text{last zero column}}.$$

$$\boxed{Av_i = \sigma_i u_i \quad \text{for every displayed column } i.}$$

Notation: Singular Values

$$\underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{rectangular diagonal matrix}}} \qquad \underbrace{\sigma_1 \geq \sigma_2 \geq \cdots \geq 0}_{\text{axis lengths}}$$

The diagonal entries are the singular values; they are lengths, so they are nonnegative.

What We Must Find

$$A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{output directions}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{nonnegative lengths}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{input-coordinate rows}}}$$

$$\underbrace{A \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{\text{contract multiplied by } V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma}$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{\text{expanded left side}} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0 u_n \\ | & | & & | \end{pmatrix}}_{\text{expanded right side}}.$$

Find the columns and the lengths; then the whole matrix is determined.

Stack the Axis Equations

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0 u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma}.$$

$$\underbrace{A \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix} \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{U\Sigma}.$$

From Axis Equations to SVD

$$\underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = I, \qquad \underbrace{V^{-1} = V^T}_{\text{orthogonal inverse}} \quad .$$

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma},$$

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}.$$

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}}.$$

Output Axes

Now Build the Output Axes

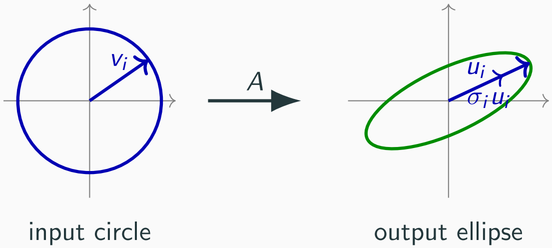
$$\underbrace{A^TA}_{\text{input-side square}} = \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{\substack{V \\ \text{input axes } v_i}} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma^T\Sigma \\ \text{squared lengths } \sigma_i^2}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T},$$

$$\underbrace{A^TAv_i = \sigma_i^2 v_i}_{\text{one input-axis column}} \qquad \underbrace{\|Av_i\|^2 = v_i^TA^TAv_i = \sigma_i^2 v_i^T v_i = \sigma_i^2}_{\text{output length computation}}.$$

$$\boxed{\underbrace{u_i}_{\text{output axis direction}} = \underbrace{\frac{Av_i}{\sigma_i}}_{\text{normalized output vector}}} \qquad \underbrace{\sigma_i > 0}_{\text{nonzero axis}}.$$

$$\boxed{\underbrace{Av_i}_{\text{actual output}} = \underbrace{\sigma_i u_i}_{\text{length times direction}}}$$

Input Axis Lands on Output Axis



v_i is an input direction. u_i is the normalized output direction.

Check u_i Has Length 1

$$\underbrace{u_i}_{\text{output unit direction}} = \underbrace{\frac{Av_i}{\sigma_i}}_{\text{normalize the output vector}} .$$

$$\|u_i\| = \left\|\frac{Av_i}{\sigma_i}\right\| = \frac{\|Av_i\|}{\sigma_i} = \frac{\sigma_i}{\sigma_i} = 1.$$

Each nonzero output axis direction is a unit vector.

Why Different Output Axes Are Orthogonal

$$\underbrace{i \neq j}_{\text{different axes}} \quad \implies \quad \underbrace{u_i^T u_j}_{\text{test orthogonality}} = \frac{(Av_i)^T (Av_j)}{\sigma_i \sigma_j}.$$

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j}.$$

$$u_i^T u_j = \frac{\sigma_j^2}{\sigma_i \sigma_j} v_i^T v_j = 0.$$

Inner-Product Table

	u_1	u_2	u_3
u_1^T	1	0	0
u_2^T	0	1	0
u_3^T	0	0	1

$$\underbrace{\begin{pmatrix} - & u_1^T & - \\ - & u_2^T & - \\ & \vdots & \\ - & u_m^T & - \end{pmatrix}}_{U^T} \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{inner-product table}} .$$

What If $\sigma_i = 0$?

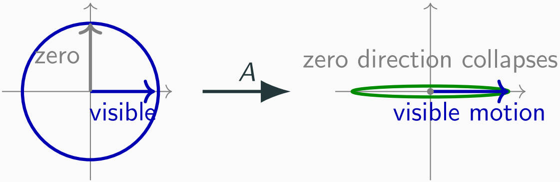
$$\underbrace{\sigma_i = 0}_{\text{zero singular value}} \quad\implies\quad \underbrace{A^T A v_i = 0}_{\text{eigenvalue } 0}.$$

$$\underbrace{\|Av_i\|^2}_{\text{squared output length}} = \underbrace{(Av_i)^T (Av_i)}_{\text{definition of length}} = \underbrace{v_i^T A^T A v_i}_{\text{move } A \text{ to the middle}} = \underbrace{v_i^T 0}_{A^T A v_i = 0} = 0.$$

$$\boxed{\underbrace{Av_i = 0}_{\text{invisible input direction}}.}$$

Zero singular value = invisible input direction.

Visible and Invisible Directions



The null space is exactly the collection of invisible directions.

Complete the Output Basis

$$\underbrace{u_1, \ldots, u_r}_{\text{directions from nonzero singular values}} \quad \rightsquigarrow \quad \underbrace{u_1, \ldots, u_r, u_{r+1}, \ldots, u_m}_{\text{full output orthonormal basis}}.$$

$$\underbrace{\mathrm{Col}(A)^{\perp}}_{\text{missing output directions}} = \underbrace{\mathrm{Null}(A^T)}_{\text{projection-complement source}}.$$

This gives a full orthogonal matrix U.

Callback: Complete by Projection Complement

$$B = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{pmatrix}}_{\text{existing output axes}}.$$

$$\underbrace{P}_{\text{projection onto known axes}} = \underbrace{BB^T}_{\text{diagonal cross-fill of known columns}}.$$

$$\underbrace{I - P}_{\text{orthogonal complement projection}}.$$

This is exactly the previous projection-complement method.

Diagonal Cross-Fill the Complement

$$\underbrace{I - P}_{\text{complement projection}} = \underbrace{w_{r+1} w_{r+1}^T + \cdots + w_m w_m^T}_{\text{rank-one orthogonal projections}}.$$

Now the missing directions come from the same old machine.

Complement Gives Missing Axes

$$\underbrace{I - P}_{\text{complement projection}} = \underbrace{w_{r+1} w_{r+1}^T + \cdots + w_m w_m^T}_{\text{diagonal cross-fill output}}.$$

$$\underbrace{u_{r+1}, \ldots, u_m}_{\text{missing output axes}} = \underbrace{w_{r+1}, \ldots, w_m}_{\text{orthonormal complement vectors}}.$$

Stack Everything

$$\underbrace{Av_i}_{\text{output from input axis}} = \underbrace{\sigma_i u_i}_{\text{axis length times output direction}}.$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma}.$$

$$\boxed{A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma}.}$$

Singular Value Decomposition

$$\underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = I, \qquad \underbrace{V^{-1} = V^T}_{\text{orthogonal inverse}} .$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma},$$

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}.$$

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}.}$$

Theorem

$$A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{orthogonal output matrix}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{nonnegative rectangular diagonal}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{orthogonal input coordinates}}}.$$

Theorem (Singular Value Decomposition)

For every real $m \times n$ matrix A, there exist orthogonal matrices U and V, and a rectangular diagonal matrix Σ with nonnegative entries, such that the displayed factorization holds.

Every real matrix maps balls to ellipsoids with orthogonal axes.

Proof Photograph

$$\underbrace{A^T A}_{\text{real symmetric}} = \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^T \Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}$$

$$\underbrace{A^T A v_i = \sigma_i^2 v_i}_{\text{input eigen-axis}} \implies \underbrace{\|Av_i\|^2 = v_i^T A^T A v_i = \sigma_i^2}_{\text{axis length}}$$

$$\underbrace{u_i = \frac{Av_i}{\sigma_i}}_{\text{normalize nonzero output}} \implies \underbrace{Av_i = \sigma_i u_i}_{\text{axis equation}}$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0 u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma}$$

$$\underbrace{AV = U\Sigma}_{\text{stacked columns}} \implies A = \underbrace{\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}}_{\text{right multiply by } V^T}.$$

The whole proof is real symmetric diagonalization of $A^T A$.

Method Summary

The Method

$$\underbrace{A^TA}_{\text{real symmetric matrix to compute}} = \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} \underbrace{\begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma^2} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}.$$

$$\underbrace{u_i = \frac{Av_i}{\sigma_i}}_{\substack{\text{define output axis} \\ \sigma_i > 0}} \quad \underbrace{B = \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{pmatrix}}_{\text{known output axes}} \quad \underbrace{P = BB^T}_{\text{known-axis projection}} \quad \underbrace{I - P}_{\text{missing-axis projection}}.$$

$$\underbrace{\begin{pmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & 0u_n \\ | & | & & | \end{pmatrix}}_{U\Sigma}$$

$$\underbrace{A\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} = U\Sigma}_{\text{stacked axis equations}} \implies A = \underbrace{\underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}}_{\text{SVD contract}}.$$

Worked Exercise

Exercise: Start With a 3×2 Matrix

$$\underbrace{A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}}_{\substack{\text{two input coordinates} \\ \text{three output coordinates}}} \qquad \underbrace{A^T = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}}_{\text{rowwise transpose}}.$$

$$\underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}}_{A^T} \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}}_{A^T A}.$$

The input-side square matrix is only 2×2.

Spectral Projections of $A^T A$

$$\underbrace{\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}}_{B=A^TA} = \underbrace{4P_4 + 0P_0}_{\text{spectral decomposition form}}, \quad \underbrace{\lambda_1 = 4,\ \lambda_2 = 0}_{\det(tI-B)=t(t-4)}.$$

$$\underbrace{P_4}_{\text{spectral projection for } \lambda=4} = \underbrace{\frac{B - 0I}{4 - 0}}_{\text{Lagrange projection polynomial}} = \underbrace{\frac{1}{4}\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}}_{\frac{1}{4}B} = \underbrace{\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_4}.$$

$$\underbrace{P_0}_{\text{spectral projection for } \lambda=0} = \underbrace{\frac{B - 4I}{0 - 4}}_{\text{Lagrange projection polynomial}} = \underbrace{\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_0}.$$

Diagonal Cross-Fill the Projections

$$\underbrace{\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_4} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{v_1 v_1^T}.$$

$$\underbrace{\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_0} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{v_2 v_2^T}.$$

$$\underbrace{v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}}_{\text{cross-filled from } P_4} \qquad \underbrace{v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}}_{\text{cross-filled from } P_0} .$$

Input Axes and Squared Singular Values

$$\underbrace{\begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{\text{orthogonal input axes}}, \quad \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}}_{\Sigma^T \Sigma}.$$

Apply A to the Input Axes

$$\underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}}_{v_1} = \underbrace{\begin{pmatrix} \sqrt{2} \\ \sqrt{2} \\ 0 \end{pmatrix}}_{Av_1} = \underbrace{2}_{\sigma_1} \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}}_{u_1}.$$

$$\underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}}_{v_2} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}}_{Av_2} = \underbrace{0}_{\sigma_2} \underbrace{u_2}_{\text{not fixed by } Av_2}.$$

Projection Onto the Known Output Axis

$$\underbrace{B = \begin{pmatrix} | \\ u_1 \\ | \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}}_{\text{known output-axis matrix}}$$

$$\underbrace{P}_{\text{projection onto known output axis}} = \underbrace{BB^T}_{\text{diagonal cross-fill}} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}}_{u_1 u_1^T} = \underbrace{\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{P}.$$

Cross-Fill the Complement $I - P$

$$\underbrace{I - P}_{\text{missing-output projection}} = \underbrace{\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{entrywise complement}} .$$

$$\underbrace{I - P}_{\text{complement projection}} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix}}_{u_2 u_2^T} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}}_{u_3 u_3^T} .$$

$$\underbrace{u_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}}_{\text{missing output axis}} \qquad \underbrace{u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{extra output axis}} .$$

Stack the Exercise Columns

$$\underbrace{A\begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix}}_{AV} = \underbrace{\begin{pmatrix} | & | \\ Av_1 & Av_2 \\ | & | \end{pmatrix}}_{\text{computed outputs}} = \underbrace{\begin{pmatrix} | & | \\ 2u_1 & 0u_2 \\ | & | \end{pmatrix}}_{\text{stretched output axes}} .$$

$$\underbrace{\begin{pmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{pmatrix}}_{U}\underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}}_{\Sigma} = \underbrace{\begin{pmatrix} | & | \\ 2u_1 & 0u_2 \\ | & | \end{pmatrix}}_{U\Sigma}.$$

Exercise Answer

$$\underbrace{U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{output orthogonal matrix}} \qquad \underbrace{\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}}_{\text{rectangular diagonal}} \qquad \underbrace{V^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{\text{input-coordinate rows}}.$$

$$\boxed{\underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{V^T}}.$$

What Each Piece Means

$$A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{orthonormal output axes}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{axis lengths of output ellipsoid}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{orthonormal input-axis coordinates}}}.$$

Input axes $\rightarrow$ stretch lengths $\rightarrow$ output axes.

The Whole Story in One Line

$$\underbrace{Av_i}_{\text{image of input axis}} = \underbrace{\sigma_i u_i}_{\text{stretched output axis}} .$$

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} .$$

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{V^T}} .$$

Final Summary

$$\boxed{\underbrace{\text{round ball}}_{\text{all unit inputs}} \xrightarrow{A} \underbrace{\text{ellipsoid with orthogonal axes}}_{\text{visible output shape}}}.$$

$$\boxed{A = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{\substack{U \\ \text{output axes}}} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}}_{\substack{\Sigma \\ \text{axis lengths}}} \underbrace{\begin{pmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}}_{\substack{V^T \\ \text{input-axis coordinates}}}}.$$

SVD records exactly this geometry.